

60-DAY DEVOPS STUDY PROGRAMME

Yacoub

1 September - 31 October 2026

Udemy "Decoding DevOps" curriculum, lectures 119-378

Combined with the in-person Tech365 programme (Tuesdays & Thursdays)

The success metric is not "I watched 378 videos".

It is: "I can build, deploy, automate, monitor, troubleshoot and EXPLAIN what I built."

1. Objectives

By 31 October 2026, Yacoub will have:

1. Completed lectures 119-378 of the Udemy course (260 lectures, 46 hours 33 minutes of video), plus his outstanding catch-up from Sections 1-13.
2. Built, broken and repaired a real multi-tier application eight separate ways - manually on EC2, through three different CI/CD tools, with Terraform, with Ansible, in Docker, and on Kubernetes.
3. A GitHub repository (yaco-devops-journey) containing eight substantial labs and one capstone, each with a README, architecture diagram, verification steps and troubleshooting notes.
4. A written troubleshooting log - the source material for every "tell me about a time something broke" interview question he will ever be asked.
5. Roughly 50 LinkedIn posts documenting real work, each with evidence attached.
6. One capstone project: a GitOps pipeline that takes a commit on GitHub all the way to a running workload on Amazon EKS, with monitoring - and the ability to explain every hop.

2. Curriculum inventory and known gaps

Everything scheduled in this document was transcribed directly from the course screenshots. Nothing has been invented. The following could not be read and is therefore NOT scheduled by lecture number:

- Lectures 1-117 (Sections 1-12 and the first part of Section 13) were not in the screenshots. Titles and durations are unknown. Because Yacoub has partially completed these, they are handled by a recurring 30-minute catch-up slot every Tuesday and Thursday at 16:00, tracked by hand in progress-tracker.md.
- Lecture 272 - title unreadable. Its duration is inferred as approximately 8 minutes from the Section 24 total.
- Lecture 139 - duration not visible; inferred as 7 minutes from the Section 14 total.
- Section 13's own title was not visible - only lectures 118-131 of it were legible.
- Lecture 118 (EBS) was already ticked complete in the screenshot, so this plan begins at lecture 119.

Note on durations: Udemy rounds each lecture to the nearest minute, so the sum of individual lectures can differ from a section header total by a few minutes. This is expected and does not indicate a transcription error.

Section	Title	Lectures	Count	Runtime
13	AWS Part-1 (tail of section - name not visible in screenshots)	118-131	14	3h 33m
14	AWS Cloud For Project Set Up Lift & Shift	132-139	8	1h 33m
15	Re-Architecting Web App on AWS Cloud [PAAS & SAAS]	140-150	11	1h 34m
16	Build tools	151-153	3	0h 45m
17	Continuous Integration and Delivery with Jenkins	154-184	31	4h 26m
18	Github Actions	185-192	8	1h 18m
19	GITLAB	193-199	7	1h 16m
20	Python	200-220	21	4h 37m
21	Learn Terraform	221-230	10	1h 38m
22	Ansible	231-249	19	4h 29m
23	Monitoring & Observability	250-262	13	2h 42m
24	AWS Part-2	263-281	19	2h 45m
25	AWS CI / CD Project	282-287	6	1h 13m
26	Project on Google Cloud	288-301	14	2h 03m

27	Docker	302-310	9	1h 59m
28	Containerization	311-323	13	2h 08m
29	Kubernetes	324-346	23	5h 06m
30	App Deployment on Kubernetes Cluster	347-360	14	1h 22m
31	Final GitOps Project	361-377	17	2h 21m
32	Conclusion	378-378	1	0h 06m

3. Does the workload fit?

Yes, with room to spare - which is the point.

The period 1 September to 31 October 2026 is 61 days. Given the schedule below (Monday/Wednesday/Friday full days, Tuesday/Thursday reduced around the Tech365 class, Saturday labs and light Sundays), Yacoub has roughly 33 usable hours per week, or about 287 hours in total.

The 260 remaining lectures total 46 hours 33 minutes of video. At the 40 percent learning / 60 percent hands-on ratio requested, that implies roughly 70 hours of practice, so about 117 hours of directly course-driven work. The remaining time goes to the eight weekly labs, documentation, troubleshooting, LinkedIn reporting, the capstone, and the Sections 1-13 catch-up.

Weekly video load ranges from 3 hours 49 minutes (Week 9) to 6 hours 30 minutes (Week 5), never more than about 2 hours on any single day. That deliberately leaves the majority of each day for building rather than watching. If Yacoub falls behind, the Tuesday and Thursday afternoons and the Sunday catch-up slots absorb the slack without redesigning the plan.

4. The running application and portfolio strategy

Every lab uses one application - the vProfile multi-tier stack from the course: an Nginx web tier, a Java/Tomcat application tier, and MySQL, Memcached and RabbitMQ behind it.

This is deliberate. Using one application throughout means each week makes the same system more sophisticated instead of starting over with a toy. By the end, Yacoub can point at one architecture and say "I deployed this by hand, then automated it, then containerised it, then orchestrated it" - which is a far stronger portfolio story than eight unrelated exercises.

Progression: manual EC2 deployment → managed services → Maven and Jenkins CI → GitHub Actions and GitLab → Terraform → Ansible → Prometheus/Grafana → custom VPC and AWS-native CI/CD → Docker → Kubernetes and Helm → GitOps capstone.

5. Weekly roadmap

Wk	Dates	Focus	Lectures	Saturday lab
1	Sep 1-6	AWS compute, storage and the Lift & Shift deployment	119-139	Lab 1: Lift & Shift: multi-tier vProfile stack on AWS EC2
2	Sep 7-13	Re-architecting on PAAS/SAAS, Maven, and Jenkins CI	140-173	Lab 2: Jenkins Continuous Integration pipeline with quality gates
3	Sep 14-20	Jenkins CD, GitHub Actions, GitLab CI, Python foundations	174-206	Lab 3: GitHub Actions CI/CD with security scanning and secrets
4	Sep 21-27	Python automation with Boto3, and Terraform	207-233	Lab 4: Terraform-provisioned infrastructure with Python/Boto3 automation
5	Sep 28-Oct 4	Ansible configuration management and observability	234-262	Lab 5: Ansible configuration management + Prometheus/Grafana observability
6	Oct 5-11	VPC networking, AWS-native CI/CD, and Google Cloud	263-301	Lab 6: Production-style VPC + AWS-native CI/CD pipeline
7	Oct 12-18	Docker and full containerization	302-323	Lab 7: Full containerization of the multi-tier application
8	Oct 19-25	Kubernetes and Helm - capstone phase 1	324-346	Lab 8: Kubernetes deployment with Helm - CAPSTONE PHASE 1
9	Oct 26-31	App on Kubernetes, GitOps, and the capstone	347-378	CAPSTONE final build & demo

Weekly milestones

Week 1 — A multi-tier application running on AWS that you built and can explain, plus a clean AWS account afterwards.

Week 2 — A Jenkinsfile in your repo that builds, tests, analyses and publishes an artifact automatically.

Week 3 — The same pipeline expressed three ways (Jenkins, GitHub Actions, GitLab) and a written comparison.

Week 4 — Infrastructure you can create and destroy with two commands, plus a Python tool that audits your own AWS account.

Week 5 — A server you never configured by hand, and a dashboard that tells you when it breaks.

Week 6 — A production-shaped private network, and an application deployed into it by a pipeline you did not touch.

Week 7 — The entire application stack running from a single `docker compose up`, with published images.

Week 8 — The application running on Kubernetes from a Helm chart, on a cluster created by Terraform.

Week 9 — A complete GitOps capstone: commit to Git, and the change reaches a live Kubernetes cluster on its own.

6. The weekly rhythm

Day	Pattern
Monday	Full learning day. 3 hours of lectures split by a break, then 3 hours hands-on and troubleshooting.

Tuesday	Tech365 class 09:00-13:00. Lighter Udemy load 14:00-15:00, short practical 15:00-16:00, Sections 1-13 catch-up 16:00-16:30.
Wednesday	Main learning day - the heaviest lecture load of the week sits here.
Thursday	Tech365 class 09:00-13:00. Same reduced pattern as Tuesday.
Friday	Learning plus integration - preparing the pieces the Saturday lab will need.
Saturday	The weekly lab. Almost entirely hands-on: build, verify, break deliberately, fix, clean up.
Sunday	Not a study day. Finish and test the lab, write the README, draw the diagram, and explain the project out loud. Finishes at 15:30.

Every study day ends with a 30-minute block (16:30-17:00, or 15:00-15:30 on Sundays) for committing to GitHub and publishing the LinkedIn report.

Weekend lectures: only four are scheduled, on Saturday 3 October (lectures 259-262, the Grafana dashboard and alerting videos). They are placed inside the lab because watching them while building the dashboard is more effective than watching them the day before. Every other Saturday and Sunday is lecture-free.

7. Detailed daily timetable

Every study day below lists the exact Udemy lecture numbers and titles to complete. Tick the boxes as you go.

Week 1 (Sep 1-6) — AWS compute, storage and the Lift & Shift deployment

Milestone: A multi-tier application running on AWS that you built and can explain, plus a clean AWS account afterwards.

Tuesday 1 September 2026

Focus: AWS storage & load balancing | Lectures 119–122 | 50 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): AWS storage & load balancing (Lectures 119–122)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Attach and snapshot an EBS volume on a t3.micro; put two EC2s behind a Classic/Application Load Balancer and prove traffic alternates.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	119	EBS Snapshots	15 min
☐	120	ELB Introduction	6 min
☐	121	ELB Hands On Part 1	10 min
☐	122	ELB Hands On Part 2	19 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 2 September 2026

Focus: CloudWatch, EFS, Autoscaling, S3 | Lectures 123–128 | 93 min of video

Time	Activity	What to do
09:00-10:30	Udemy: CloudWatch, EFS, Autoscaling, S3 (Lectures 123–127)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: CloudWatch, EFS, Autoscaling, S3 (Lectures 128)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Build a CloudWatch alarm on CPU, mount an EFS share across two instances, create a launch template + ASG (min 1 / max 2), create your first S3 bucket.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the

		cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	123	Cloudwatch Introduction	5 min
☐	124	Cloudwatch Hands On	19 min
☐	125	EFS	18 min
☐	126	Autoscaling Group Introduction	4 min
☐	127	Autoscaling Group Hands On	25 min
☐	128	S3 Introduction	22 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 3 September 2026

Focus: S3 static hosting & RDS | Lectures 129–131 | 49 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): S3 static hosting & RDS (Lectures 129–131)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Host a one-page site on S3, then launch a free-tier db.t3.micro MySQL RDS instance and connect to it from an EC2 over a security group rule.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	129	S3 Website Hosting	12 min
☐	130	More in S3	11 min
☐	131	RDS	26 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 4 September 2026

Focus: Lift & Shift project setup | Lectures 132–139 | 93 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Lift & Shift project setup (Lectures 132–135)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Lift & Shift project setup (Lectures 136–139)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.

12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Walk the whole Section 14 stack once, following the lectures exactly. Take notes on every security group rule - you will rebuild it unaided tomorrow.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	132	Introduction	11 min
☐	133	Security Group & Keypairs	10 min
☐	134	EC2 Instances	19 min
☐	135	DNS Route 53	7 min
☐	136	Build and Deploy Artifacts	16 min
☐	137	Load Balancer & DNS	11 min
☐	138	Autoscaling Group	12 min
☐	139	Validate & Summarize	7 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 5 September 2026

Time	Activity	What to do
09:00-11:00	LAB 1: Lift & Shift: multi-tier vProfile stack on AWS EC2 - build phase	Rebuild the Section 14 architecture from scratch WITHOUT replaying the videos: MySQL, Memcached and RabbitMQ on private EC2s, Tomcat app EC2, Nginx web EC2, an Application Load Balancer in front, a Route 53 record, and an Autoscaling Group for the app tier. Everything free-tier sized.
11:00-11:15	Break	
11:15-13:00	LAB 1: Lift & Shift: multi-tier vProfile stack on AWS EC2 - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 1: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 6 September 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the	Finish and document Lab 1. Draw the lift-and-shift architecture

	week's lab	by hand, photograph or redraw it digitally, and record a 5-minute walkthrough.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos
 ☐ Exercise
 ☐ Lab work
 ☐ GitHub updated
 ☐ LinkedIn posted
 ☐ Presentation

Week 2 (Sep 7-13) — Re-architecting on PAAS/SAAS, Maven, and Jenkins CI

Milestone: A Jenkinsfile in your repo that builds, tests, analyses and publishes an artifact automatically.

Monday 7 September 2026

Focus: Re-architecting on PAAS/SAAS | Lectures 140–150 | 94 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Re-architecting on PAAS/SAAS (Lectures 140–146)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Re-architecting on PAAS/SAAS (Lectures 147–150)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Deploy the app to Elastic Beanstalk backed by RDS, ElastiCache and Amazon MQ. Front it with CloudFront. DESTROY EVERYTHING before you close the laptop.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	140	Introduction	13 min
☐	141	Security Group And Keypairs	4 min
☐	142	RDS	11 min
☐	143	Elastic Cache	5 min
☐	144	Amazon MQ	3 min
☐	145	DB Initialization	7 min
☐	146	Beanstalk	19 min
☐	147	Update on Security Group & ELB	2 min
☐	148	Build & Deploy Artifact	13 min
☐	149	Cloud front	9 min
☐	150	Validate and Summarize	8 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 8 September 2026

Focus: Build tools - Maven | Lectures 151–153 | 45 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Build tools - Maven (Lectures 151–153)	Lighter load today by design. Quality of attention beats volume.

15:00-16:00	Short practical reinforcement	Clone the vProfile source, run `mvn clean install`, find the produced .war in target/ and explain what each Maven lifecycle phase did.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	151	Introduction	9 min
☐	152	Maven Hands-on	24 min
☐	153	Maven, NodeJS & AI	12 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 9 September 2026

Focus: Jenkins install & first jobs | Lectures 154-161 | 72 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Jenkins install & first jobs (Lectures 154-158)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Jenkins install & first jobs (Lectures 159-161)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Install Jenkins on an Ubuntu EC2, add JDK + Maven as global tools, create a freestyle job that builds the vProfile war, then add a build agent.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	154	Introduction	6 min
☐	155	Installation	13 min
☐	156	Freestyle Vs Pipeline As A Code	2 min
☐	157	Installing tools in Jenkins	7 min
☐	158	First Job	8 min
☐	159	First Build Job	11 min
☐	160	Agents	13 min
☐	161	Plugins, Versioning & more	12 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 10 September 2026

Focus: CI flow, Nexus & SonarQube setup | Lectures 162–166 | 24 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): CI flow, Nexus & SonarQube setup (Lectures 162–166)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Stand up Nexus and SonarQube servers. Keep them t3.small or smaller and STOP them when you finish.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	162	Disk Space Issue	3 min
☐	163	Flow of Continuous Integration Pipeline	5 min
☐	164	Steps for Continuous Integration Pipeline	2 min
☐	165	Jenkins, Nexus & Sonarqube Setup	13 min
☐	166	Plugins for CI	1 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 11 September 2026

Focus: Pipeline as code, quality gates, Slack | Lectures 167–173 | 72 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Pipeline as code, quality gates, Slack (Lectures 167–169)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Pipeline as code, quality gates, Slack (Lectures 170–173)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Write your first Jenkinsfile: Build -> Test -> Code Analysis -> Quality Gate -> Upload artifact to Nexus -> Slack notification.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	167	Pipeline As A Code Introduction	18 min
☐	168	Code Analysis	7 min

☐	169	Code Analysis Demonstration	14 min
☐	170	Quality Gates	7 min
☐	171	Software Repositories Intro (Nexus)	5 min
☐	172	Nexus PAAC Demo	9 min
☐	173	Notification, Slack	12 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 12 September 2026

Time	Activity	What to do
09:00-11:00	LAB 2: Jenkins Continuous Integration pipeline with quality gates - build phase	Build a complete CI pipeline as code for the vProfile app: checkout, Maven build, unit tests, Checkstyle, SonarQube analysis with a quality gate that can FAIL the build, artifact upload to Nexus, and a Slack notification on both success and failure.
11:00-11:15	Break	
11:15-13:00	LAB 2: Jenkins Continuous Integration pipeline with quality gates - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 2: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 13 September 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish and document Lab 2. Record a screen capture of the pipeline running green, and a second one of it failing the quality gate on purpose.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted ☐ Presentation

Week 3 (Sep 14-20) — Jenkins CD, GitHub Actions, GitLab CI, Python foundations

Milestone: The same pipeline expressed three ways (Jenkins, GitHub Actions, GitLab) and a written comparison.

Monday 14 September 2026

Focus: Jenkins CI/CD to Docker & ECS | Lectures 174–184 | 98 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Jenkins CI/CD to Docker & ECS (Lectures 174–180)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Jenkins CI/CD to Docker & ECS (Lectures 181–184)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Extend the Jenkinsfile to build a Docker image and push it, then deploy to ECS. Configure a webhook build trigger and lock the instance down with matrix authorisation.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	174	CI for Docker Intro	2 min
☐	175	Docker PAAC Prereqs info	9 min
☐	176	Docker PAAC Demo	15 min
☐	177	Docker CICD Intro	4 min
☐	178	Docker CICD Code	2 min
☐	179	AWS ECS Setup	13 min
☐	180	Docker CICD Demonstration	5 min
☐	181	Cleanup	2 min
☐	182	Build Triggers Intro	14 min
☐	183	Build Triggers Demo	18 min
☐	184	Authentication & Authorization	14 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 15 September 2026

Focus: GitHub Actions quickstart | Lectures 185–188 | 43 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): GitHub Actions quickstart (Lectures 185–188)	Lighter load today by design. Quality of attention beats volume.

15:00-16:00	Short practical reinforcement	Write <code>.github/workflows/ci.yml</code> that runs on push to a feature branch and on <code>pull_request</code> , and prove both triggers fire.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in <code>progress-tracker.md</code> .
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	185	Introduction	5 min
☐	186	Quickstart Part1	16 min
☐	187	Quickstart Part2	9 min
☐	188	Triggers and Inputs	13 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 16 September 2026

Focus: Actions artifacts/security + GitLab start | Lectures 189-195 | 63 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Actions artifacts/security + GitLab start (Lectures 189-192)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Actions artifacts/security + GitLab start (Lectures 193-195)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Add an artifact upload and a security scan job to the Actions workflow. Then create a GitLab account and run your first <code>.gitlab-ci.yml</code> .
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in <code>docs/troubleshooting-log.md</code> .
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using <code>linkedin-report-template.md</code> .

Lectures to complete:

☐	189	Artifacts, Conditions & Repo Permissions	8 min
☐	190	Security Scan	7 min
☐	191	Secrets & Docker	7 min
☐	192	Build & Publish Job	13 min
☐	193	Introduction	5 min
☐	194	Initial Setup	13 min
☐	195	First Pipeline	10 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 17 September 2026

Focus: GitLab variables, rules, Docker | Lectures 196–199 | 48 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): GitLab variables, rules, Docker (Lectures 196–199)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Move a hard-coded value into a masked GitLab CI/CD variable, and add a rule so the deploy stage only runs on the main branch.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	196	Variables & More	10 min
☐	197	Triggers & Rules	12 min
☐	198	Security & Artifacts	15 min
☐	199	Build And Publish Docker	11 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 18 September 2026

Focus: Python foundations for DevOps | Lectures 200–206 | 77 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Python foundations for DevOps (Lectures 200–203)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Python foundations for DevOps (Lectures 204–206)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Write and run five small scripts covering variables, string slicing, operators and conditions. Commit them to labs/python/.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	200	Introduction	10 min
☐	201	Python on Linux, Versions & Indentation	10 min
☐	202	Quotes and Comments	4 min

☐	203	Variables	15 min
☐	204	Print Format	5 min
☐	205	Slicing	16 min
☐	206	Operators	17 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 19 September 2026

Time	Activity	What to do
09:00-11:00	LAB 3: GitHub Actions CI/CD with security scanning and secrets - build phase	Port the Jenkins pipeline to GitHub Actions, then reproduce the core of it a third time in GitLab CI. Build the app, scan it, build and publish a Docker image, and gate deployment behind a branch rule. Write a short comparison of the three tools.
11:00-11:15	Break	
11:15-13:00	LAB 3: GitHub Actions CI/CD with security scanning and secrets - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 3: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 20 September 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish and document Lab 3. Write the Jenkins vs GitHub Actions vs GitLab comparison properly - this is a common interview question.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted ☐ Presentation

Week 4 (Sep 21-27) — Python automation with Boto3, and Terraform

Milestone: Infrastructure you can create and destroy with two commands, plus a Python tool that audits your own AWS account.

Monday 21 September 2026

Focus: Python control flow, functions, modules | Lectures 207–213 | 93 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Python control flow, functions, modules (Lectures 207–210)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Python control flow, functions, modules (Lectures 211–213)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Refactor yesterday's five scripts into one module with functions. No copy-paste duplication allowed.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	207	Conditions	15 min
☐	208	Loops	14 min
☐	209	Break & Continue	12 min
☐	210	Built-in Functions or Methods	17 min
☐	211	Functions part-1	17 min
☐	212	Functions part-2	11 min
☐	213	Modules	7 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 22 September 2026

Focus: OS tasks & Python Fabric | Lectures 214–215 | 55 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): OS tasks & Python Fabric (Lectures 214–215)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Write a script that reads /var/log, filters error lines and writes a summary file. Then use Fabric to run `uptime` on a remote EC2.
16:00-16:30	Catch-up: unfinished lectures from	Reserved slot for the earlier part of the course you have not

	Sections 1-13	finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	214	OS Tasks	23 min
☐	215	Python Fabric	32 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 23 September 2026

Focus: Boto3, AI-assisted automation, Terraform basics | Lectures 216–223 | 92 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Boto3, AI-assisted automation, Terraform basics (Lectures 216–219)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Boto3, AI-assisted automation, Terraform basics (Lectures 220–223)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Write a Boto3 script that lists every running EC2 with its tags and flags any without an Owner tag. Then write your first main.tf.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	216	Exception Handling	10 min
☐	217	Cloud Interaction with Boto3	11 min
☐	218	AI for Cloud Automation	17 min
☐	219	Copilot AI for Cloud Automation	14 min
☐	220	Python Scripts	0 min
☐	221	Introduction	7 min
☐	222	Basics of Terraform	16 min
☐	223	Code Structure	17 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 24 September 2026

Focus: Terraform code structure & variables | Lectures 224–226 | 36 min of video

Time	Activity	What to do
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09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemmy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemmy (light): Terraform code structure & variables (Lectures 224-226)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Split your main.tf into main.tf / variables.tf / outputs.tf and parameterise the AMI and instance type.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	224	Code Structure Part 2	6 min
☐	225	Plan, Apply, Update & Destroy	17 min
☐	226	Variables	13 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 25 September 2026

Focus: Provisioners, outputs, backend + Ansible intro | Lectures 227-233 | 63 min of video

Time	Activity	What to do
09:00-10:30	Udemmy: Provisioners, outputs, backend + Ansible intro (Lectures 227-231)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemmy: Provisioners, outputs, backend + Ansible intro (Lectures 232-233)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Move Terraform state to an S3 backend. Then install Ansible and ping two Terraform-created EC2s from an inventory file.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	227	Provisioners	10 min
☐	228	Outputs	5 min
☐	229	Backend	5 min
☐	230	What Next?	2 min

☐	231	Introduction	16 min
☐	232	Setup Ansible & Infra	11 min
☐	233	Inventory & Ping Module	14 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 26 September 2026

Time	Activity	What to do
09:00-11:00	LAB 4: Terraform-provisioned infrastructure with Python/Boto3 automation - build phase	Replace all manual clicking with code. Write Terraform that provisions the app tier (EC2s, security groups, key pair, S3 bucket, remote state backend) and a Python/Boto3 tool that audits the account: lists running instances, flags untagged resources, and reports estimated monthly cost of what is running.
11:00-11:15	Break	
11:15-13:00	LAB 4: Terraform-provisioned infrastructure with Python/Boto3 automation - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 4: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 27 September 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish and document Lab 4. Run `terraform destroy`, then run your Boto3 auditor and paste the empty output into the README as proof of cleanup.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted ☐ Presentation

Week 5 (Sep 28-Oct 4) — Ansible configuration management and observability

Milestone: A server you never configured by hand, and a dashboard that tells you when it breaks.

Monday 28 September 2026

Focus: Ansible inventory, ad hoc, playbooks | Lectures 234–239 | 92 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Ansible inventory, ad hoc, playbooks (Lectures 234–237)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Ansible inventory, ad hoc, playbooks (Lectures 238–239)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Write a playbook that installs Nginx, copies a custom index.html and starts the service on both hosts. Run it twice and explain idempotency.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	234	Inventory Part 2	19 min
☐	235	YAML & JSON	9 min
☐	236	Ad Hoc Commands	12 min
☐	237	Playbook & Modules	21 min
☐	238	Modules - Find, Use, Troubleshoot & Repeat	20 min
☐	239	Ansible Configuration	11 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 29 September 2026

Focus: Ansible variables & facts | Lectures 240–242 | 40 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Ansible variables & facts (Lectures 240–242)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Convert hard-coded values in your playbook into group_vars, then print three fact variables with the debug module.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing

		built.
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Lectures to complete:

☐	240	Variables & Debug	14 min
☐	241	Group & Host Variables	15 min
☐	242	Fact Variables	11 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 30 September 2026

Focus: Conditionals, loops, templates, roles, AWS | Lectures 243–248 | 96 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Conditionals, loops, templates, roles, AWS (Lectures 243–247)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Conditionals, loops, templates, roles, AWS (Lectures 248)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Refactor the Nginx playbook into a proper role (tasks/handlers/templates/vars) and template the config file with Jinja2.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	243	Decision Making	12 min
☐	244	Loops	6 min
☐	245	File, copy & template modules	20 min
☐	246	Handlers	7 min
☐	247	Roles	31 min
☐	248	Ansible for AWS	20 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 1 October 2026

Focus: vProfile Ansible code + monitoring intro | Lectures 249–252 | 31 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): vProfile Ansible code +	Lighter load today by design. Quality of attention beats volume.

	monitoring intro (Lectures 249–252)	
15:00-16:00	Short practical reinforcement	Read the vProfile Ansible code and map each role to something you already built by hand in Week 1.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	249	Vprofile code	0 min
☐	250	Introduction to Monitoring	15 min
☐	251	Why Monitoring is Essential for DevOps	3 min
☐	252	Monitoring and Observability Tools	13 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 2 October 2026

Focus: Prometheus, node exporter, PromQL, Grafana | Lectures 253–258 | 78 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Prometheus, node exporter, PromQL, Grafana (Lectures 253–255)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Prometheus, node exporter, PromQL, Grafana (Lectures 256–258)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Install Prometheus + node_exporter on one EC2 and Grafana alongside it. Query node_cpu_seconds_total in PromQL and connect Grafana.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	253	Setting Up the Monitoring Environment	15 min
☐	254	Loki and Web Server Setup	20 min
☐	255	Adding a Node to Prometheus	16 min
☐	256	Understanding PromQL (Prometheus Query Language)	16 min
☐	257	Connecting Grafana & Prometheus	7 min
☐	258	Integrating Slack for Notifications	4 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 3 October 2026

Focus: Grafana dashboards, alerts, Loki | Lectures 259–262 | 53 min of video

Time	Activity	What to do
09:00-10:00	Udemy: Grafana dashboards, alerts, Loki (Lectures 259–262)	SATURDAY LECTURES (justified): these four are the dashboard/alerting build itself - watching them inside the lab is more effective than watching them a day early.
10:00-13:00	LAB 5: Ansible configuration management + Prometheus/Grafana observability - build phase	Provision infrastructure with Terraform, configure it entirely with Ansible roles, and monitor it with Prometheus, Grafana, node_exporter and Loki. Build a real dashboard and a real alert that fires to Slack.
13:00-14:00	Lunch	
14:00-15:45	LAB 5: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

Lectures to complete:

☐	259	PromQL for Grafana Dashboards	12 min
☐	260	Building Grafana Panels and Dashboards	17 min
☐	261	Creating Alerts and Thresholds	11 min
☐	262	Integrating Loki and Alloy for Logs and Metrics	13 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 4 October 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish and document Lab 5. Export your Grafana dashboard JSON into the repo so anyone can import it. Mid-point review: are you on track?
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted ☐ Presentation

Week 6 (Oct 5-11) — VPC networking, AWS-native CI/CD, and Google Cloud

Milestone: A production-shaped private network, and an application deployed into it by a pipeline you did not touch.

Monday 5 October 2026

Focus: VPC design and components | Lectures 263–272 | 77 min of video

Time	Activity	What to do
09:00-10:30	Udemy: VPC design and components (Lectures 263–265)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: VPC design and components (Lectures 266–272)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Draw the target VPC on paper first: 2 public + 2 private subnets, IGW, NAT, route tables. Then build it in the console.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	263	VPC Introduction	28 min
☐	264	VPC Design & Components	9 min
☐	265	VPC Setup Details	4 min
☐	266	Default VPC	7 min
☐	267	Create VPC	6 min
☐	268	Subnets	3 min
☐	269	Internet Gateway	2 min
☐	270	Route Tables	4 min
☐	271	NAT Gateway	6 min
☐	272	[TITLE NOT READABLE IN SCREENSHOT]	8 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 6 October 2026

Focus: Website in VPC, peering, Terraform VPC | Lectures 273–275 | 38 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Website in VPC, peering, Terraform VPC (Lectures 273–275)	Lighter load today by design. Quality of attention beats volume.

15:00-16:00	Short practical reinforcement	Recreate today's VPC in Terraform. Compare the console version with the code version - which one could you rebuild in 5 minutes?
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	273	Website in VPC	14 min
☐	274	Peering	11 min
☐	275	Terraform for VPC Setup	13 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 7 October 2026

Focus: EC2 logs, Lambda, AWS CI/CD project | Lectures 276-287 | 123 min of video

Time	Activity	What to do
09:00-10:30	Udemy: EC2 logs, Lambda, AWS CI/CD project (Lectures 276-284)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: EC2 logs, Lambda, AWS CI/CD project (Lectures 285-287)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Build the CodeCommit -> CodeBuild -> CodePipeline -> Beanstalk pipeline end to end. Delete the pipeline and Beanstalk env afterwards.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	276	Ec2 Logs	31 min
☐	277	AWS Lambda	19 min
☐	278	Links	0 min
☐	279	buildspec	0 min
☐	280	Links	0 min
☐	281	S3 policy	0 min
☐	282	Introduction	4 min
☐	283	Beanstalk	7 min
☐	284	RDS & App Setup on Beanstalk	12 min

☐	285	Code Commit	14 min
☐	286	Code build	22 min
☐	287	Build, Deploy & Code Pipeline	14 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 8 October 2026

Focus: GCP vProfile project intro | Lectures 288–291 | 30 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): GCP vProfile project intro (Lectures 288–291)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Create a GCP free-trial project, install gcloud, and set your project/region defaults. Do not enable billing-heavy APIs you do not need.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	288	Introduction to the GCP vProfile Project	11 min
☐	289	Project Architecture Overview	7 min
☐	290	Setting Up Your GCP Account & Project	6 min
☐	291	Commands in the Source Code	6 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 9 October 2026

Focus: GCP network, Cloud SQL, MIG, LB, DNS | Lectures 292–301 | 93 min of video

Time	Activity	What to do
09:00-10:30	Udemy: GCP network, Cloud SQL, MIG, LB, DNS (Lectures 292–296)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: GCP network, Cloud SQL, MIG, LB, DNS (Lectures 297–301)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Complete the GCP deployment and then run the Section 301 cleanup in full. Verify in the billing console that nothing is still running.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.

16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.
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Lectures to complete:

☐	292	Configuring Project Variables	7 min
☐	293	VPC, Subnets & Network Setup	9 min
☐	294	Firewall Rules & VM Deployment	14 min
☐	295	Configuring Cloud SQL & Memorystore	11 min
☐	296	Setting Up Cloud DNS	8 min
☐	297	Creating a Custom VM Image	11 min
☐	298	Building a Managed Instance Group (MIG)	5 min
☐	299	Configuring the Global HTTP/HTTPS Load Balancer	11 min
☐	300	SSL Certificates, HTTPS & Final DNS Setup	9 min
☐	301	Summary and Cleanup	8 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 10 October 2026

Time	Activity	What to do
09:00-11:00	LAB 6: Production-style VPC + AWS-native CI/CD pipeline - build phase	Build a proper three-tier VPC in Terraform (public/private subnets across two AZs, IGW, NAT gateway, route tables, bastion) and deploy the application into it through an AWS-native CodeCommit/CodeBuild/CodePipeline flow.
11:00-11:15	Break	
11:15-13:00	LAB 6: Production-style VPC + AWS-native CI/CD pipeline - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 6: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 11 October 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish and document Lab 6. Redraw the VPC diagram from memory without looking at your notes - if you cannot, you have not learned it yet.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	

14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos
 ☐ Exercise
 ☐ Lab work
 ☐ GitHub updated
 ☐ LinkedIn posted
 ☐ Presentation

Week 7 (Oct 12-18) — Docker and full containerization

Milestone: The entire application stack running from a single `docker compose up`, with published images.

Monday 12 October 2026

Focus: Docker engine, commands, logs, volumes | Lectures 302–306 | 66 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Docker engine, commands, logs, volumes (Lectures 302–304)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Docker engine, commands, logs, volumes (Lectures 305–306)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Run Nginx in a container, break it deliberately, read the logs, then attach a named volume and prove data survives a container delete.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	302	Introduction	10 min
☐	303	Docker Setup	9 min
☐	304	Docker commands & concepts	22 min
☐	305	Docker Logs	8 min
☐	306	Docker volumes	17 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 13 October 2026

Focus: Building images, ENTRYPOINT vs CMD | Lectures 307–308 | 28 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Building images, ENTRYPOINT vs CMD (Lectures 307–308)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Write a Dockerfile for a tiny app. Change ENTRYPOINT to CMD and back, and write down in your notes what actually changed.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing

		built.
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Lectures to complete:

☐	307	Building images	21 min
☐	308	Entrypoint and CMD	7 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 14 October 2026

Focus: Compose, multi-stage, base images, Dockerhub | Lectures 309-315 | 60 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Compose, multi-stage, base images, Dockerhub (Lectures 309-311)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Compose, multi-stage, base images, Dockerhub (Lectures 312-315)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Write a multi-stage Dockerfile that builds the war in one stage and copies only the artifact into a slim runtime stage. Compare image sizes.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	309	Docker Compose	15 min
☐	310	Multi Stage Dockerfile	10 min
☐	311	Introduction	10 min
☐	312	Overview of Base Image	9 min
☐	313	Dockerhub Setup	3 min
☐	314	Setup Docker Engine	6 min
☐	315	Dockerhub & Dockerfile References	7 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 15 October 2026

Focus: App, DB and Web image Dockerfiles | Lectures 316-318 | 23 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	

14:00-15:00	Udemy (light): App, DB and Web image Dockerfiles (Lectures 316–318)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Build all three vProfile images locally and push them to your Docker Hub account under your own namespace.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	316	App Image Dockerfile	10 min
☐	317	DB Image Dockerfile	7 min
☐	318	Web Image Dockerfile	6 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 16 October 2026

Focus: Compose stack + microservice containerization | Lectures 319–323 | 70 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Compose stack + microservice containerization (Lectures 319–321)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Compose stack + microservice containerization (Lectures 322–323)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Bring the entire stack up with one <code>`docker compose up -d`</code> , then containerize the microservice project.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	319	Docker Compose	19 min
☐	320	Build and Run	11 min
☐	321	Summarize	6 min
☐	322	Containerizing Microservice Project	22 min
☐	323	Build & Run Microservice App	12 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 17 October 2026

Time	Activity	What to do
09:00-11:00	LAB 7: Full containerization of the multi-tier application - build phase	Containerize every tier: a multi-stage Dockerfile for the Java app, a custom DB image with the schema pre-loaded, an Nginx web image, plus Memcached and RabbitMQ. Bring the whole thing up with one docker compose command and publish the images.
11:00-11:15	Break	
11:15-13:00	LAB 7: Full containerization of the multi-tier application - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 7: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 18 October 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish and document Lab 7. Then WRITE THE CAPSTONE PLAN: architecture diagram, repo layout, and the order you will build it in next week.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted ☐ Presentation

Week 8 (Oct 19-25) — Kubernetes and Helm - capstone phase 1

Milestone: The application running on Kubernetes from a Helm chart, on a cluster created by Terraform.

Monday 19 October 2026

Focus: Kubernetes intro, Minikube, Kops, kubeconfig | Lectures 324–328 | 70 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Kubernetes intro, Minikube, Kops, kubeconfig (Lectures 324–326)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Kubernetes intro, Minikube, Kops, kubeconfig (Lectures 327–328)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Get a working Minikube cluster and run <code>kubectl get nodes</code> . Understand your kubeconfig before touching a cloud cluster.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	324	Introduction	23 min
☐	325	Minikube for K8s Setup	9 min
☐	326	Kops for K8s Setup	23 min
☐	327	Objects and Documentation	5 min
☐	328	Kube Config	10 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 20 October 2026

Focus: Pods, namespaces, logging | Lectures 329–331 | 28 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Pods, namespaces, logging (Lectures 329–331)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Create a pod from YAML, put it in its own namespace, then deliberately mistype the image name and read the resulting event stream.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	329	Pods	11 min
☐	330	Namespace	8 min
☐	331	Different levels of Logging	9 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 21 October 2026

Focus: Services, ReplicaSets, Deployments, ConfigMaps | Lectures 332–337 | 75 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Services, ReplicaSets, Deployments, ConfigMaps (Lectures 332–334)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Services, ReplicaSets, Deployments, ConfigMaps (Lectures 335–337)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Deploy 3 replicas behind a ClusterIP service, roll out an image change, then roll it back with `kubectl rollout undo`.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	332	Service	21 min
☐	333	Replica Set	11 min
☐	334	Deployment	12 min
☐	335	Command and Arguments	7 min
☐	336	Volumes	9 min
☐	337	Config Map	15 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 22 October 2026

Focus: Secrets and Ingress | Lectures 338–339 | 31 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Secrets and Ingress (Lectures 338–339)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	Create a secret from a literal, mount it as an env var, and expose

		a deployment through an Ingress.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	338	Secret	12 min
☐	339	Ingress	19 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 23 October 2026

Focus: *kubectl CLI, Helm, Lens, Terraform for EKS* | Lectures 340–346 | 102 min of video

Time	Activity	What to do
09:00-10:30	Udemy: kubectl CLI, Helm, Lens, Terraform for EKS (Lectures 340–343)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: kubectl CLI, Helm, Lens, Terraform for EKS (Lectures 344–346)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Package one of your deployments as a Helm chart with values.yaml. Read (do not yet apply) the Terraform EKS module.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	340	Kubectl CLI & Cheatsheet	12 min
☐	341	Extras	13 min
☐	342	Helm Introduction	6 min
☐	343	Helm Hands On	20 min
☐	344	Helm with AI	19 min
☐	345	Lens	7 min
☐	346	Terraform For EKS Setup	25 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 24 October 2026

Time	Activity	What to do
09:00-11:00	LAB 8: Kubernetes deployment with Helm - CAPSTONE PHASE 1 - build	This lab is the first phase of your capstone. Provision an EKS cluster with Terraform (or run on Minikube if you want to avoid

	phase	EKS cost), package the application as a Helm chart, and deploy it with secrets, persistent volumes, services and an Ingress.
11:00-11:15	Break	
11:15-13:00	LAB 8: Kubernetes deployment with Helm - CAPSTONE PHASE 1 - continue	Keep going. Note every error message you hit.
13:00-14:00	Lunch	
14:00-15:45	LAB 8: verification & deliberate breakage	Verify against the acceptance criteria in the lab README, then work the troubleshooting challenges.
15:45-16:30	Cleanup & cost check	Destroy or stop every billable resource. Check the AWS Billing dashboard before you stop.
16:30-17:00	Commit + LinkedIn lab report	Today's post should describe what you BUILT, with a screenshot as evidence.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Sunday 25 October 2026

Time	Activity	What to do
10:00-12:00	Finish, test and troubleshoot the week's lab	Finish Lab 8 / capstone phase 1. Confirm the cluster comes up from `terraform apply` alone. Prepare the capstone build checklist for the final week.
12:00-13:00	Documentation: README, diagram, screenshots	Update the lab README: objective, architecture, steps, verification, troubleshooting, cleanup.
13:00-14:00	Lunch / rest	
14:00-15:00	Explain the week's project out loud	Answer all six questions aloud and record it: What did I build? Why? How does it work? What broke? How did I troubleshoot it? What would I improve?
15:00-15:30	Weekly reflection + LinkedIn summary post	One weekly post that ties the week together, with your best evidence attached.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted ☐ Presentation

Week 9 (Oct 26-31) — App on Kubernetes, GitOps, and the capstone

Milestone: A complete GitOps capstone: commit to Git, and the change reaches a live Kubernetes cluster on its own.

Monday 26 October 2026

Focus: vProfile on Kubernetes - secrets, PVC, services | Lectures 347–356 | 56 min of video

Time	Activity	What to do
09:00-10:30	Udemy: vProfile on Kubernetes - secrets, PVC, services (Lectures 347–352)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: vProfile on Kubernetes - secrets, PVC, services (Lectures 353–356)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	Apply the full manifest set to your cluster and debug until every pod is Running. Expect failures - that is the point.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	347	Introduction	5 min
☐	348	Architecture	7 min
☐	349	Source Code Overview	5 min
☐	350	Secret	5 min
☐	351	Persistent Volume for DB [PVC]	4 min
☐	352	MySQL App	10 min
☐	353	MySQL Service	4 min
☐	354	Memcache App & Service	4 min
☐	355	RabbitMQ App & Service	5 min
☐	356	Tomcat App & Service	7 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Tuesday 27 October 2026

Focus: Ingress and full cluster deployment | Lectures 357–360 | 26 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): Ingress and full cluster	Lighter load today by design. Quality of attention beats volume.

	deployment (Lectures 357–360)	
15:00-16:00	Short practical reinforcement	Reach the app in a browser through the Ingress. Screenshot it - this goes in the capstone README.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	357	Ingress	6 min
☐	358	K8s Cluster Setup & Source Code	4 min
☐	359	Deploy App on K8s Cluster	12 min
☐	360	Summarize	4 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Wednesday 28 October 2026

Focus: *GitOps project: Helm charts, CI pipeline, ECR* | Lectures 361–371 | 77 min of video

Time	Activity	What to do
09:00-10:30	Udemy: GitOps project: Helm charts, CI pipeline, ECR (Lectures 361–367)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: GitOps project: Helm charts, CI pipeline, ECR (Lectures 368–371)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	CAPSTONE BUILD: repo structure, Helm charts, SonarQube, ECR + IAM, GitHub secrets, CI/CD pipeline parts 1 and 2.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.

Lectures to complete:

☐	361	Introduction	6 min
☐	362	Architecture	6 min
☐	363	Git Repository Setup	6 min
☐	364	Helm Charts Part 1	3 min
☐	365	Helm Charts Part 2	13 min
☐	366	CI Pipeline Overview	3 min
☐	367	Sonar Qube Server	6 min
☐	368	ECR and IAM	6 min

☐	369	Github Secrets & Variables	6 min
☐	370	CICD Pipeline part 1	12 min
☐	371	CICD Pipeline part 2	10 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Thursday 29 October 2026

Focus: EKS prereqs, Terraform EKS, Argo CD | Lectures 372–375 | 45 min of video

Time	Activity	What to do
09:00-13:00	Tech365 in-person DevOps class	Fixed commitment - no Udemy work during this block.
13:00-14:00	Lunch and reset	
14:00-15:00	Udemy (light): EKS prereqs, Terraform EKS, Argo CD (Lectures 372–375)	Lighter load today by design. Quality of attention beats volume.
15:00-16:00	Short practical reinforcement	CAPSTONE BUILD: provision EKS with Terraform, install Argo CD, connect it to your Helm chart repo and watch it sync.
16:00-16:30	Catch-up: unfinished lectures from Sections 1-13	Reserved slot for the earlier part of the course you have not finished. Work through it in order and tick it off in progress-tracker.md.
16:30-17:00	Commit to GitHub + LinkedIn report	Short post is fine on Tech365 days - one thing learned, one thing built.

Lectures to complete:

☐	372	EKS Prereqs	5 min
☐	373	Terraform Code for EKS	10 min
☐	374	Argo CD Setup	11 min
☐	375	Argo App Sync	19 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Friday 30 October 2026

Focus: Completing the setup, summary, resumes | Lectures 376–378 | 25 min of video

Time	Activity	What to do
09:00-10:30	Udemy: Completing the setup, summary, resumes (Lectures 376–377)	Watch actively: pause after each lecture and write one sentence in your notes about what it does and when you would use it.
10:30-10:45	Break	Stand up, leave the screen.
10:45-12:15	Udemy: Completing the setup, summary, resumes (Lectures 378)	Same rule: no passive watching. If a lecture is a demo, follow along in your own terminal.
12:15-13:15	Lunch	
13:15-15:15	Hands-on practice	CAPSTONE BUILD: finish the end-to-end flow, then update your CV with the project while it is fresh.
15:15-15:30	Break	
15:30-16:30	Troubleshooting & revision	Break something on purpose, then fix it. Write the symptom, the cause and the fix in docs/troubleshooting-log.md.

16:30-17:00	Commit to GitHub + LinkedIn report	Push today's work, then publish the LinkedIn post using linkedin-report-template.md.
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Lectures to complete:

☐	376	Completing the whole Setup	10 min
☐	377	Summary	9 min
☐	378	Resumes	6 min

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

Saturday 31 October 2026

Time	Activity	What to do
09:00-11:00	CAPSTONE: final end-to-end run	Destroy everything and rebuild the entire capstone from your own code and README. If it does not come up, your documentation is not finished.
11:00-11:15	Break	
11:15-13:00	CAPSTONE: monitoring, screenshots, diagram	Grafana dashboard live, Argo CD sync screenshot, architecture diagram final version.
13:00-14:00	Lunch	
14:00-15:30	CAPSTONE: README, demo script, interview answers	Write the demo instructions so a stranger can reproduce it. Answer all six explanation questions in writing.
15:30-16:30	Final presentation - record it	Record a 10-15 minute walkthrough of the whole two months. This is your portfolio centrepiece.
16:30-17:00	Final LinkedIn post + tear down all cloud resources	Post the capstone. Then run terraform destroy everywhere and confirm a zero-resource account.

☐ Videos ☐ Exercise ☐ Lab work ☐ GitHub updated ☐ LinkedIn posted

8. The eight weekly labs

Each Saturday lab integrates that week's concepts and builds on the previous one. Full instructions, verification steps, troubleshooting challenges, cleanup commands and interview questions live in the repository under `labs/lab-NN/README.md`.

Lab 1: Lift & Shift: multi-tier vProfile stack on AWS EC2

Saturday 5 September 2026

Rebuild the Section 14 architecture from scratch WITHOUT replaying the videos: MySQL, Memcached and RabbitMQ on private EC2s, Tomcat app EC2, Nginx web EC2, an Application Load Balancer in front, a Route 53 record, and an Autoscaling Group for the app tier. Everything free-tier sized.

Acceptance criteria:

- ☐ Security group chained rules (web -> app -> db) rather than 0.0.0.0/0 everywhere
- ☐ Artifact build with Maven and deploy to Tomcat
- ☐ Route 53 private hosted zone for backend service discovery
- ☐ Health checks that actually fail when the app is down

Troubleshooting challenges (hints only, no answers):

- Your ALB target group shows 'unhealthy' but the app runs fine when you curl it from the instance itself.
- Tomcat starts but the app page returns 500 with a database connection error.
- The Route 53 name resolves on one instance but not another.

Lab 2: Jenkins Continuous Integration pipeline with quality gates

Saturday 12 September 2026

Build a complete CI pipeline as code for the vProfile app: checkout, Maven build, unit tests, Checkstyle, SonarQube analysis with a quality gate that can FAIL the build, artifact upload to Nexus, and a Slack notification on both success and failure.

Acceptance criteria:

- ☐ Jenkinsfile committed to the repo, not a freestyle job
- ☐ SonarQube quality gate wired through the webhook
- ☐ Nexus versioned artifact upload
- ☐ Credentials stored in Jenkins credentials store, never in the Jenkinsfile

Troubleshooting challenges (hints only, no answers):

- The pipeline hangs forever at the Quality Gate stage.
- Maven works on your laptop but the Jenkins agent cannot find JDK 11.
- The Nexus upload returns 400 Bad Request.

Lab 3: GitHub Actions CI/CD with security scanning and secrets

Saturday 19 September 2026

Port the Jenkins pipeline to GitHub Actions, then reproduce the core of it a third time in GitLab CI. Build the app, scan it, build and publish a Docker image, and gate deployment behind a branch rule. Write a short comparison of the three tools.

Acceptance criteria:

- ☐ Matrix or multi-job workflow with dependencies between jobs
- ☐ GitHub Secrets for registry credentials - nothing in plaintext
- ☐ A security scan job that can fail the workflow
- ☐ A written Jenkins vs Actions vs GitLab comparison in the lab README

Troubleshooting challenges (hints only, no answers):

- The workflow passes locally with `act` but fails in GitHub with a permissions error.
- The Docker push fails with 'denied: requested access to the resource is denied'.

- The deploy job runs on a feature branch even though you added an `if` condition.

Lab 4: Terraform-provisioned infrastructure with Python/Boto3 automation

Saturday 26 September 2026

Replace all manual clicking with code. Write Terraform that provisions the app tier (EC2s, security groups, key pair, S3 bucket, remote state backend) and a Python/Boto3 tool that audits the account: lists running instances, flags untagged resources, and reports estimated monthly cost of what is running.

Acceptance criteria:

- ☐ Remote state in S3 with a DynamoDB lock table
- ☐ Variables and outputs - no hard-coded AMIs or CIDRs
- ☐ A `terraform destroy` that leaves the account genuinely clean
- ☐ The Boto3 auditor script run before and after destroy, with output saved

Troubleshooting challenges (hints only, no answers):

- `terraform apply` fails with a dependency cycle.
- State is locked and you cannot apply - the previous run crashed.
- Boto3 raises NoCredentialsError inside a script that worked in the shell.

Lab 5: Ansible configuration management + Prometheus/Grafana observability

Saturday 3 October 2026

Provision infrastructure with Terraform, configure it entirely with Ansible roles, and monitor it with Prometheus, Grafana, node_exporter and Loki. Build a real dashboard and a real alert that fires to Slack.

Acceptance criteria:

- ☐ Ansible roles with handlers and Jinja2 templates - no long flat playbooks
- ☐ Prometheus scraping at least two nodes
- ☐ A Grafana dashboard with CPU, memory, disk and app-availability panels
- ☐ An alert rule with a threshold you can trigger on demand

Troubleshooting challenges (hints only, no answers):

- Ansible reports 'changed' on every run - your playbook is not idempotent.
- Prometheus shows the target as DOWN but you can curl :9100/metrics from the Prometheus host.
- Grafana panels are empty even though the datasource test succeeds.

Lab 6: Production-style VPC + AWS-native CI/CD pipeline

Saturday 10 October 2026

Build a proper three-tier VPC in Terraform (public/private subnets across two AZs, IGW, NAT gateway, route tables, bastion) and deploy the application into it through an AWS-native CodeCommit/CodeBuild/CodePipeline flow.

Acceptance criteria:

- ☐ Application and database tiers in PRIVATE subnets with no public IPs
- ☐ NAT gateway for egress only - note that this one DOES cost money hourly
- ☐ buildspec.yml committed to the repo
- ☐ A network diagram you drew yourself, committed as PNG

Troubleshooting challenges (hints only, no answers):

- Private instances cannot reach the internet to install packages.
- CodeBuild fails with 'no matching artifact' at the end of the build.
- You can SSH to the bastion but not from the bastion to the app server.

Lab 7: Full containerization of the multi-tier application

Saturday 17 October 2026

Containerize every tier: a multi-stage Dockerfile for the Java app, a custom DB image with the schema pre-loaded, an Nginx web image, plus Memcached and RabbitMQ. Bring the whole thing up with one docker compose command and publish the images.

Acceptance criteria:

- ☐ Multi-stage build with a measurably smaller final image
- ☐ docker-compose.yml with named volumes, a custom network and healthchecks
- ☐ Images pushed to Docker Hub under your own account
- ☐ A .dockerignore that actually excludes build junk

Troubleshooting challenges (hints only, no answers):

- The app container starts then exits immediately with code 0.
- The app cannot resolve the hostname 'mysql' even though both containers are running.
- Data disappears every time you run `docker compose down`.

Lab 8: Kubernetes deployment with Helm - CAPSTONE PHASE 1

Saturday 24 October 2026

This lab is the first phase of your capstone. Provision an EKS cluster with Terraform (or run on Minikube if you want to avoid EKS cost), package the application as a Helm chart, and deploy it with secrets, persistent volumes, services and an Ingress.

Acceptance criteria:

- ☐ Terraform-managed cluster, not a console-created one
- ☐ One Helm chart with a values.yaml that switches between dev and prod settings
- ☐ Secrets NOT committed - use kubectl create secret or an external secret reference
- ☐ A working Ingress you can reach in a browser

Troubleshooting challenges (hints only, no answers):

- Pods stuck in Pending - describe the pod and read the events.
- CrashLoopBackOff on the app pod with a database connection refused error.
- The PVC stays Pending forever on your cluster.

9. The capstone project

GitOps deployment of a multi-tier application to Kubernetes on AWS

Built 26-31 October 2026. In one sentence: "I push a commit to GitHub, and a few minutes later that change is running on a Kubernetes cluster on AWS, with monitoring, without anyone running a deploy command."

Architecture

Developer pushes to GitHub

- GitHub Actions: Maven build → unit tests → SonarQube quality gate → multi-stage Docker build
- Image pushed to Amazon ECR; the Helm chart image tag is updated in Git
- Argo CD detects the change in Git and syncs it
- Amazon EKS (provisioned by Terraform, workers in private subnets) runs the workload
- Nginx Ingress → app Deployment → MySQL (PVC), Memcached, RabbitMQ
- Prometheus scrapes it, Grafana displays it, alerts route to Slack

What each week contributes

Week	Skill	Where it appears
1	AWS fundamentals, security groups	EKS node group networking
2	Maven, Jenkins CI, quality gates	The build and analysis stages
3	GitHub Actions, secrets management	The entire CI pipeline
4	Terraform, Python/Boto3	EKS and VPC provisioning, cost auditing
5	Ansible, Prometheus, Grafana	The monitoring stack and alerting
6	VPC design, private subnets	Worker nodes in private subnets
7	Docker, multi-stage builds	The application image in ECR
8	Kubernetes, Helm	The chart Argo CD syncs
9	GitOps, Argo CD	The deployment mechanism itself

Deliverables

- ☐ Architecture diagram drawn by Yacoub, not copied
- ☐ Application source code repository
- ☐ Terraform for VPC, EKS, ECR and IAM - destroyable in one command
- ☐ GitHub Actions CI/CD pipeline
- ☐ Helm chart and Argo CD application manifest
- ☐ Prometheus and Grafana deployed, dashboard JSON committed
- ☐ README with reproduction instructions a stranger could follow
- ☐ Troubleshooting notes covering everything that went wrong
- ☐ Screenshots: pipeline green, Argo CD synced, app in a browser, Grafana dashboard
- ☐ A recorded 10-15 minute walkthrough
- ☐ terraform destroy run and verified

Cost warning

EKS bills per cluster-hour, plus worker nodes, plus a NAT gateway, plus EBS volumes for persistent volume claims. This is the most expensive item in the whole programme. Build it in a focused session, capture the screenshots, then destroy it. Rebuilding from his own Terraform is a feature, not a chore - it proves the code works.

10. Cloud cost safety

Rule zero: set an AWS billing alarm before launching anything. Billing → Budgets → monthly budget with an email alert, set low (five dollars is plenty).

Resource	Why it costs	What to do
NAT Gateway	Hourly charge plus data processing, no free tier	Delete the moment the lab ends
Elastic Load Balancer	Free tier is limited hours only	Delete after every lab
Unattached Elastic IP	Charged precisely because it is idle	Release it
EKS control plane	Flat hourly charge per cluster	Use Minikube where possible; destroy EKS same-day
EBS volumes and snapshots	Storage charges accumulate silently	Delete PVCs after helm uninstall
RDS Multi-AZ	Doubles the cost, outside free tier	Single-AZ, db.t3.micro only
Route 53 hosted zones	Small monthly charge each	Delete when the lab is finished
GCP Cloud SQL / Global LB	Outside most free tiers	Run the Section 301 cleanup in full

End-of-session sweep (run it every single day, not every week):

```
terraform destroy -auto-approve
terraform state list                # must return nothing
python3 scripts/aws_audit.py --all-regions
aws ec2 describe-nat-gateways --query 'NatGateways[?
State==`available`].NatGatewayId'
aws ec2 describe-addresses --query 'Addresses[].PublicIp'
aws eks list-clusters
```

11. Secrets management

No AWS key, database password, API token, kubeconfig or private key ever enters the repository. Not once, not temporarily, not in a branch that will be deleted.

Where secrets belong	Instead of
AWS CLI profile or IAM role	Access keys pasted into a .tf file
Jenkins credentials store, referenced by ID	A password written into the Jenkinsfile
GitHub Actions repository secrets	A token in the workflow YAML
GitLab masked CI/CD variables	A variable committed to .gitlab-ci.yml
Ansible Vault (encrypted files are safe to commit)	Plaintext passwords in group_vars
kubectl create secret, from a shell variable	A Secret manifest committed to Git - base64 is not encryption
A .env file listed in .gitignore	Values hard-coded in docker-compose.yml

Pre-commit habit: run `grep -riE "password|secret|token|AKIA"` . before every push. If something does leak, rotate the credential first and clean the history second - it is compromised from the moment it reaches GitHub.

12. Daily LinkedIn report

Scheduled 16:30-17:00 on study days, 15:00-15:30 on Sundays. Consistency beats brilliance: forty honest ordinary posts beat three polished ones.

Template

Day X of my 60-day DevOps journey.

Today I learned: <the concept, in your own words>

Today I built: <what you actually made - be concrete>

The problem I hit: <the real error message or symptom>

How I solved it: <what you checked, what the cause turned out to be>

What I now understand that I didn't yesterday: <the insight>

Never post "Today I watched videos 154-161." Nobody can hire from that. Always attach evidence: a post without a screenshot is a claim, a post with one is proof. Write about the failure, not just the success - engineers respect the debugging story.

Evidence to attach

Topic	Attach
Linux / Bash	Terminal screenshot with the command and its output
AWS	Console screenshot, or better, your own architecture diagram
Jenkins	Pipeline stage view - one green, one failing the quality gate
GitHub Actions	Workflow run summary page
Terraform	terraform plan output, and the resulting resources
Ansible	PLAY RECAP showing changed=0 on the second run
Docker	docker images showing your multi-stage size saving
Kubernetes	kubectl get pods -o wide, or the app through the Ingress
Prometheus / Grafana	The dashboard, with a real alert firing
Any week	The GitHub commit or pull request

Four worked example posts - including one about a day that went badly - are in `linkedin-report-template.md` in the repository.

13. Progress tracker

Weekly milestones

- ☐ Week 1 (Sep 1-6) — A multi-tier application running on AWS that you built and can explain, plus a clean AWS account afterwards.
- ☐ Week 2 (Sep 7-13) — A Jenkinsfile in your repo that builds, tests, analyses and publishes an artifact automatically.
- ☐ Week 3 (Sep 14-20) — The same pipeline expressed three ways (Jenkins, GitHub Actions, GitLab) and a written comparison.
- ☐ Week 4 (Sep 21-27) — Infrastructure you can create and destroy with two commands, plus a Python tool that audits your own AWS account.
- ☐ Week 5 (Sep 28-Oct 4) — A server you never configured by hand, and a dashboard that tells you when it breaks.
- ☐ Week 6 (Oct 5-11) — A production-shaped private network, and an application deployed into it by a pipeline you did not touch.
- ☐ Week 7 (Oct 12-18) — The entire application stack running from a single `docker compose up`, with published images.
- ☐ Week 8 (Oct 19-25) — The application running on Kubernetes from a Helm chart, on a cluster created by Terraform.
- ☐ Week 9 (Oct 26-31) — A complete GitOps capstone: commit to Git, and the change reaches a live Kubernetes cluster on its own.

Sections 1-13 catch-up (lectures 1-117)

These were not visible in the screenshots, so they cannot be listed by number here. Track them by hand in the Tuesday and Thursday 16:00-16:30 slot.

- ☐ Section _____ complete (lectures _____ to _____)
- ☐ Section _____ complete (lectures _____ to _____)
- ☐ Section _____ complete (lectures _____ to _____)
- ☐ Section _____ complete (lectures _____ to _____)
- ☐ Section _____ complete (lectures _____ to _____)
- ☐ Section _____ complete (lectures _____ to _____)

Weekly self-check

Every Sunday, answer these six questions out loud about the week's lab. If any answer is vague, that is the thing to revisit before moving on.

- ☐ What did I build?
- ☐ Why did I build it?
- ☐ How does it work?
- ☐ What problem did I encounter?
- ☐ How did I troubleshoot it?
- ☐ What would I improve?

A full day-by-day tracker with checkboxes for every one of the 61 days is in progress-tracker.md in the repository.

14. Repository structure

yaco-devops-journey/	
├─ README.md	Journey overview and weekly roadmap
├─ progress-tracker.md	All 61 days with checkboxes
├─ linkedin-report-template.md	Template plus four worked examples
├─ .gitignore	Blocks keys, state, .env, kubeconfig
├─ docs/	Study plan, curriculum inventory, cost safety, troubleshooting log
├─ architecture/	One diagram per lab plus the capstone
├─ week-01/ ... week-09/	Daily notes
├─ labs/lab-01/ ... lab-08/	Objective, architecture, steps, verification, troubleshooting, cleanup, interview questions
├─ capstone/	Terraform, Helm, Argo CD, docs
├─ terraform/	Root stack, vpc/, eks/
├─ ansible/	ansible.cfg, site.yml, roles/nginx
├─ docker/	Three Dockerfiles, compose, .dockerignore
├─ kubernetes/	Manifests, Helm chart, SECRETS.md
├─ jenkins/	Jenkinsfile with quality gate and Slack
├─ github-actions/	ci.yml with scan and conditional publish
├─ monitoring/	prometheus.yml, alert-rules.yml
├─ scripts/	aws_audit.py, health_check.sh

Starter files already written and ready to extend include: Terraform for a tagged EC2 stack and a two-AZ VPC with NAT; an idempotent Ansible nginx role with handlers and a Jinja2 template; three Dockerfiles including a multi-stage build; a docker-compose stack with healthchecks and named volumes; Kubernetes manifests with probes, resource limits and secret references; a Helm chart; a Jenkinsfile with a blocking quality gate; a GitHub Actions workflow with a security scan and branch-conditional publish; Prometheus scrape and alert rules; a Python cost auditor; and a Bash tier health checker.

15. The weekly cycle

LEARN → PRACTISE → BUILD → BREAK IT → TROUBLESHOOT → DOCUMENT → COMMIT → EXPLAIN

Week 1 is heavily guided: every command is explained, every step is spelled out. By Weeks 7 and 8 the labs give objectives and acceptance criteria and expect Yacoub to work out the how. The troubleshooting challenges never include the answer - only a hint - because the ability to sit with a broken system and reason about it is the actual job.

If a week slips, do not skip the lab to catch up on videos. Skip videos to protect the lab. A finished lab he can explain is worth more than a section he watched at 2x speed.